

Higher-Order Structures and Diverse Functions

Intended Learning Outcomes

1. Differentiate between the primary, secondary, tertiary, and quaternary levels of protein architecture, specifying the chemical bonds that stabilize each level.
2. Describe the diverse physiological tasks of proteins in biological systems, including active transport, structural support, enzymatic catalysis, and signaling.
3. Evaluate how environmental factors induce protein denaturation

From Amino Acid Sequence to Functional Protein

They perform many functions, such as catalysis, molecular transport, DNA replication, and structural support.

Primary Structure



amino acid
sequence

Secondary Structure

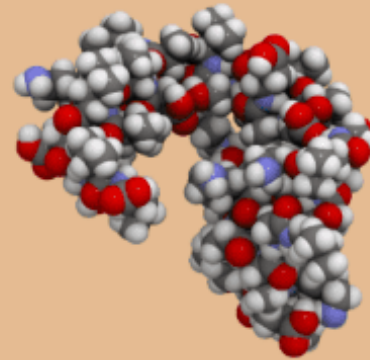
Alpha helix



Pleated sheet

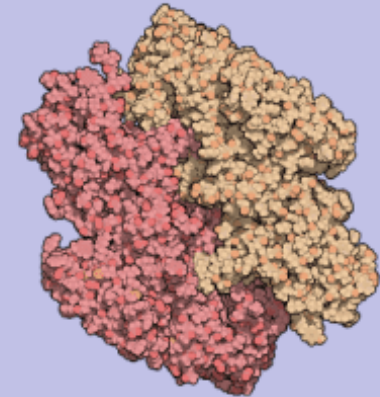
sequence folds
into 3D shape

Tertiary Structure



mature protein
folds upon itself

Quaternary Structure



protein consists of
multiple
polypeptide chains

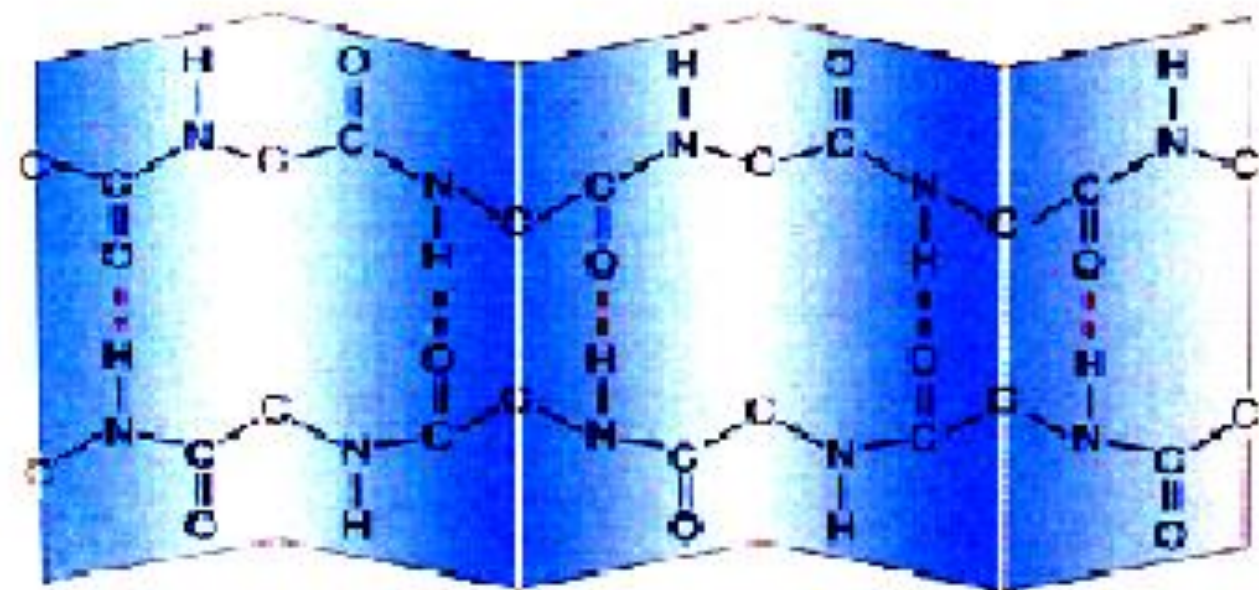
From Amino Acid Sequence to Functional Protein

- **Primary Structure:** The linear sequence of amino acids.
- **Secondary Structure:** Localized, recurring structures (like alpha-helices and beta-sheets) formed by hydrogen bonding within the polypeptide backbone.
- **Tertiary Structure:** The overall 3D shape of a single polypeptide chain, resulting from interactions between amino acid side chains.
- **Quaternary Structure:** The arrangement of multiple polypeptide chains (subunits) in a multi-subunit protein.
- Each level of structure is built upon the previous one, and the final 3D shape, known as the **conformation**, is essential for a protein to perform its specific biological role.

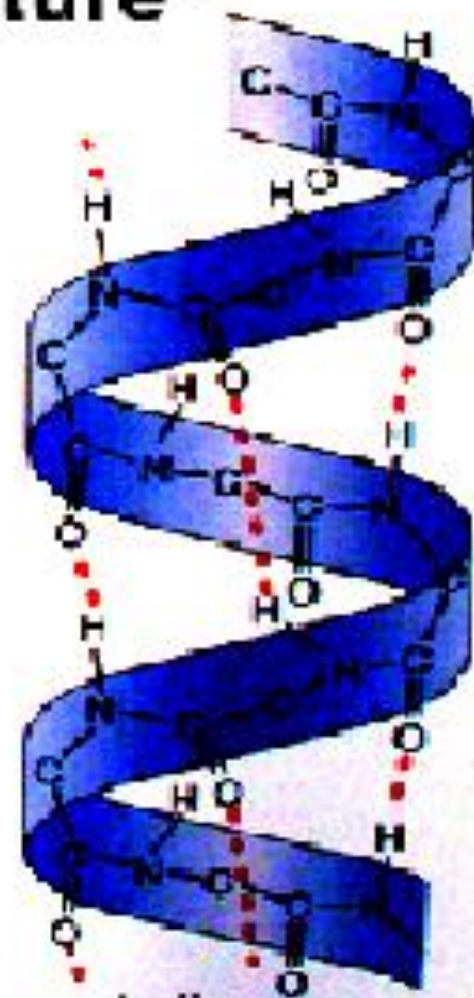
Secondary Structure: Localized Folding Patterns

- The polypeptide backbone, despite its rigidity at the peptide bond, has flexibility around the α -carbons.
- This flexibility, combined with hydrogen bonding between atoms of the polypeptide backbone (not the R-groups), gives rise to common, repeating local structures called secondary structures.

secondary structure



β pleated sheet



α helix

The Alpha (α)-Helix

- **Description:** A common helical (spiral) structure, resembling a coiled spring. The polypeptide chain coils around an imaginary central axis.
- **Stabilization:** Stabilized by **hydrogen bonds** formed between the **carbonyl oxygen** of one peptide bond and the **amide hydrogen** of a peptide bond **four amino acids away** along the polypeptide chain.
- **Location:** Found extensively in many proteins, both globular and fibrous. For example, the protein keratin (in hair and nails) is rich in α -helices.

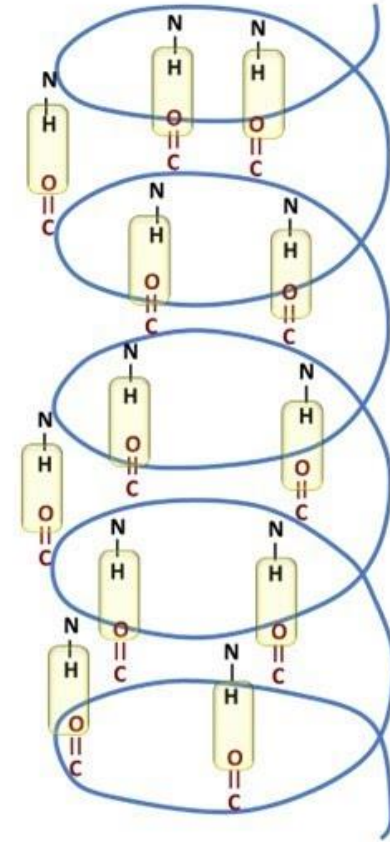
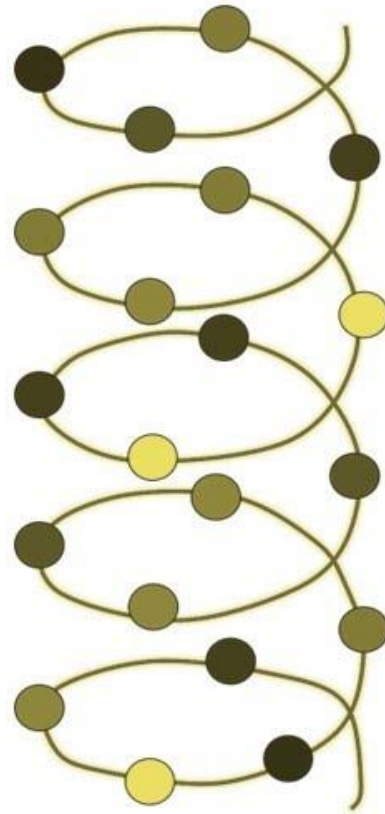
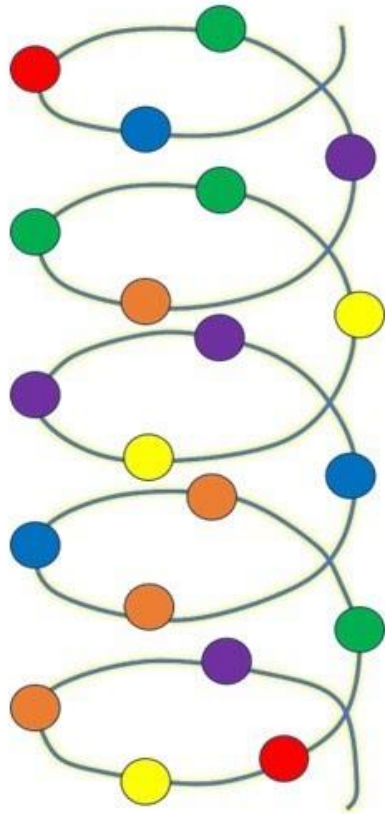
The Alpha (α)-Helix

- **Key Features**

- **Right-handed helix:** Almost all biological alpha-helices are right-handed.
- **Hydrogen bonds parallel to axis:** The hydrogen bonds run roughly parallel to the long axis of the helix.
- **R-groups point outwards:** The amino acid side chains (R-groups) project outwards from the helix, minimizing steric hindrance and allowing them to interact with the environment or other parts of the protein.
- **Residues per turn:** Approximately 3.6 amino acid residues per turn.
- **Pitch:** The rise per turn is 5.4 Å (angstroms).

Quickly
Understand

α Helix



The Beta (β)-Sheet

- **Description:** Consists of two or more polypeptide strands (called β -strands) running alongside each other, connected by hydrogen bonds. It has a pleated, zig-zag appearance.
- **Stabilization:** Stabilized by **hydrogen bonds** formed between the **carbonyl oxygens** and **amide hydrogens** of peptide bonds on **adjacent β -strands**.
 - These hydrogen bonds are perpendicular to the polypeptide chain direction.
- **Location:** Abundant in many globular proteins. For example, the fatty acid-binding proteins use β -sheets to form a "barrel" structure.

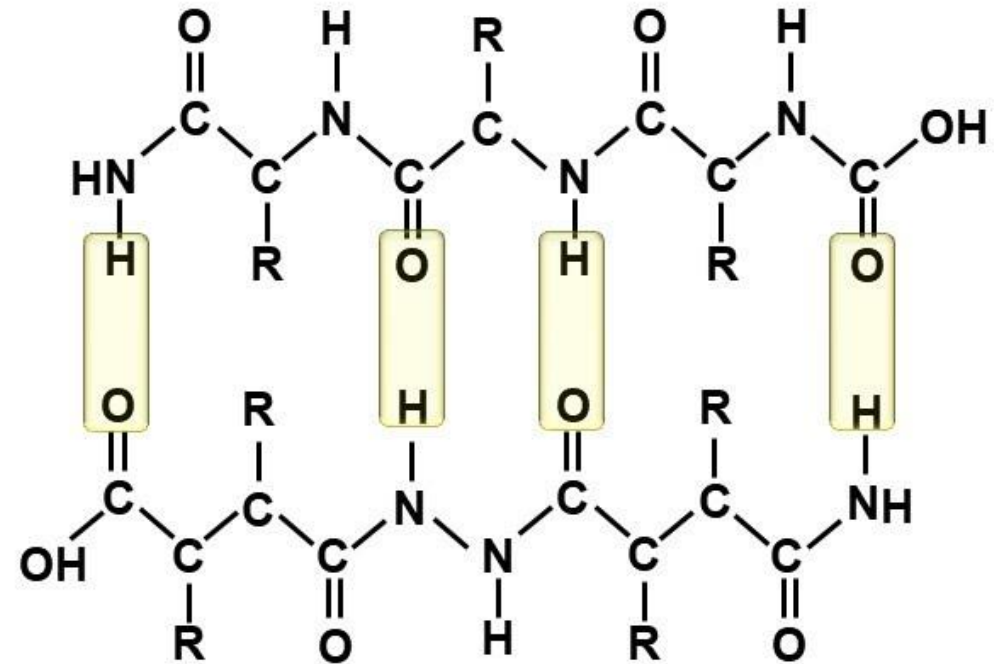
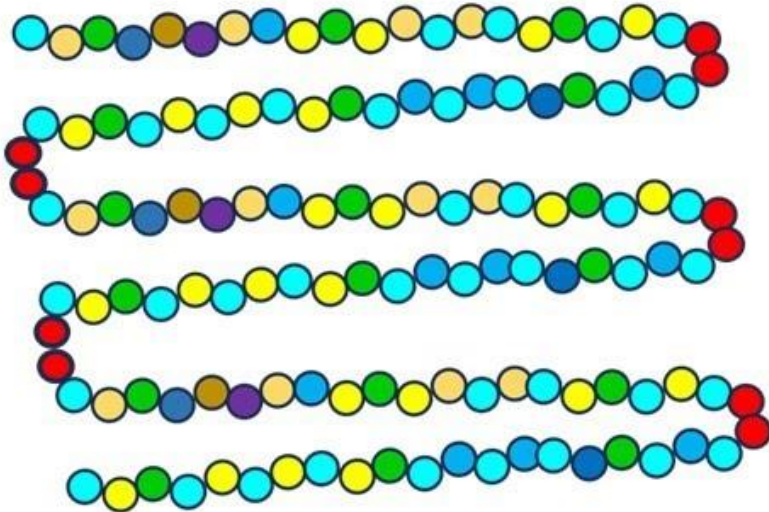
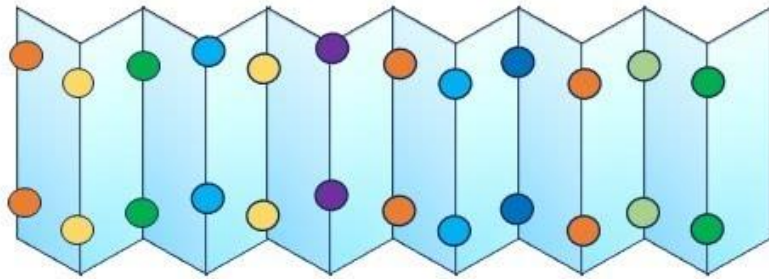
The Beta (β)-Sheet

- **Key Features**

- **Flatness:** β -sheets are generally flatter and more extended than α -helices.
- **R-groups alternate:** Amino acid side chains (R-groups) project alternately above and below the plane of the sheet.
- **Parallel vs. Antiparallel:**
 - **Antiparallel β -sheet:** Adjacent strands run in opposite N-terminal to C-terminal directions. The hydrogen bonds are perfectly aligned and stronger. More common and stable.
 - **Parallel β -sheet:** Adjacent strands run in the same N-terminal to C-terminal direction. The hydrogen bonds are somewhat distorted and weaker. Less common.

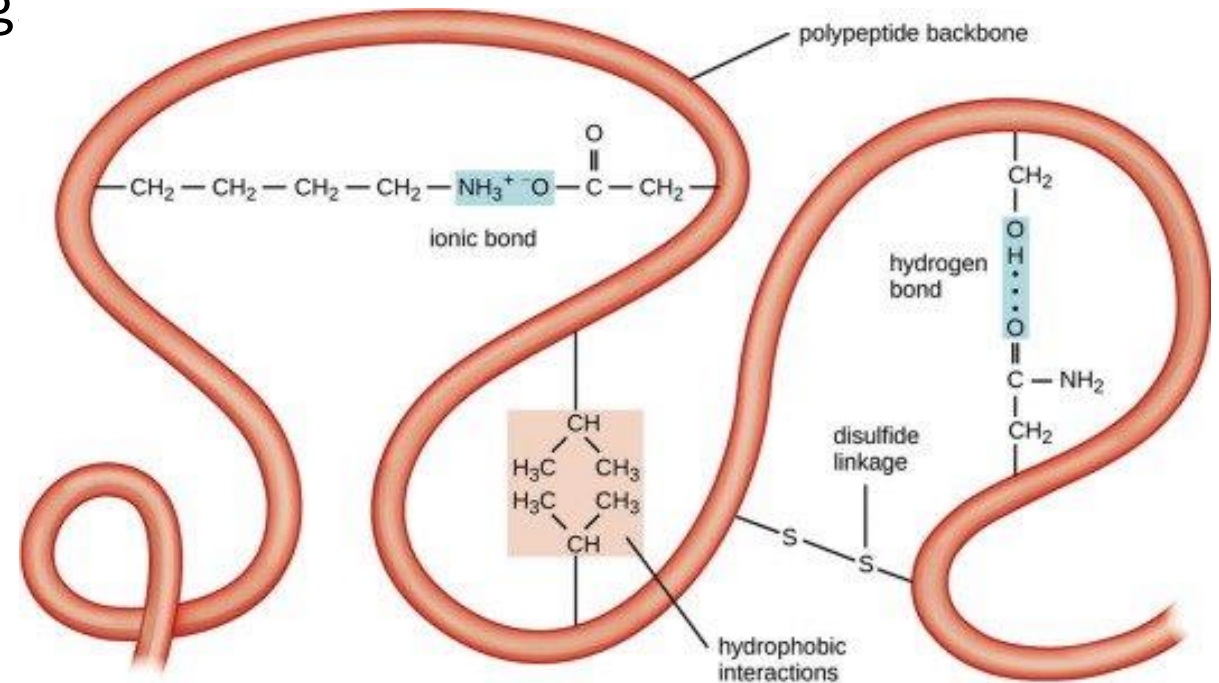
Quickly
Understand

β Pleated sheet



Tertiary Structure: The Overall 3D Fold

- The tertiary structure is the complete, three-dimensional arrangement of all atoms in a single polypeptide chain.
- It is the result of intricate interactions between the **side chains (R-groups)** of amino acids that are often far apart in the primary sequence but are brought close together by secondary folding



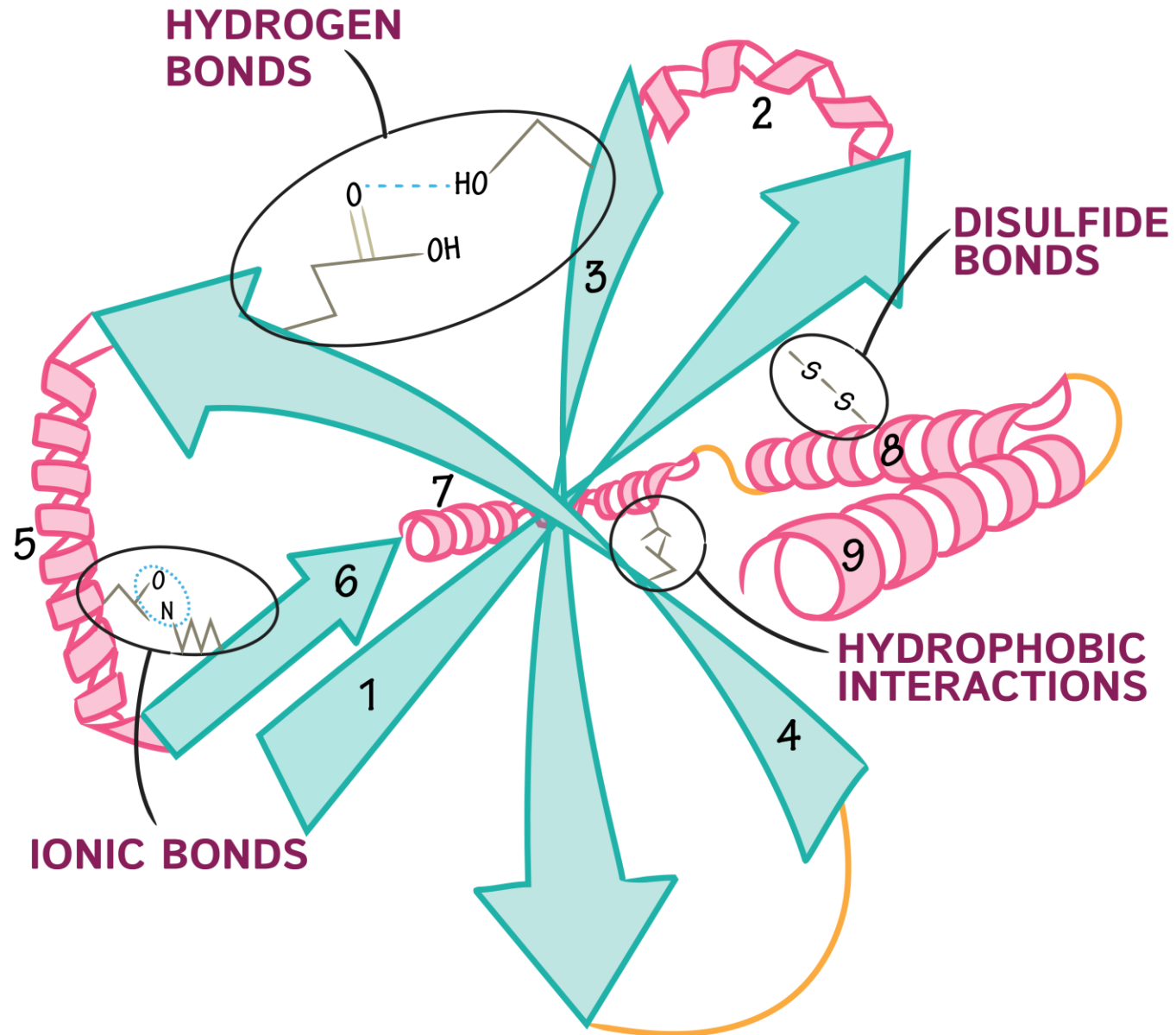
Forces Stabilizing Tertiary Structure

- **Hydrophobic Interactions (Major Driving Force):** Nonpolar R-groups tend to cluster together in the interior of the protein, away from the aqueous environment of the cytoplasm. This minimizes contact with water and is a major driving force for protein folding.
- **Hydrogen Bonds:** Can form between polar R-groups, or between polar R-groups and the polypeptide backbone.

Forces Stabilizing Tertiary Structure

- **Ionic Bonds (Salt Bridges):** electrostatic interactions between oppositely charged R-groups (e.g., between a positively charged Lysine and a negatively charged Aspartate).
- **Disulfide Bonds (Covalent Bonds):** Strong covalent bonds formed by the oxidation of two cysteine residues. These are particularly important for stabilizing the structure of secreted proteins or proteins in oxidizing environments (e.g., antibodies).
- **Van der Waals Interactions:** Weak, transient attractions between all atoms that are in close proximity. Summing up many such interactions can contribute significantly to stability.

Tertiary Structure



Protein Domains

- Within larger proteins, tertiary structure can often be organized into independent compact globular units called **domains**.
- Each domain typically folds independently and may have a specific function (e.g., a binding domain, a catalytic domain).
- Proteins with multiple functions often have multiple domains.

Globular vs. Fibrous Proteins

Globular Proteins

- **Shape:** Roughly spherical or compact.
- **Solubility:** Generally water-soluble (hydrophilic R-groups on the surface, hydrophobic in the core).
- **Function:** Diverse functions, including enzymes, transporters, receptors, immunoglobulins (antibodies), and many hormones. (e.g., Hemoglobin, Lysozyme).

Fibrous Proteins

- **Shape:** Elongated, rod-like, often insoluble.
- **Solubility:** Generally water-insoluble.
- **Function:** Structural roles, providing strength, flexibility, or elasticity. Often form long fibers or sheets. (e.g., Collagen, Keratin, Silk Fibroin).

Protein Folding

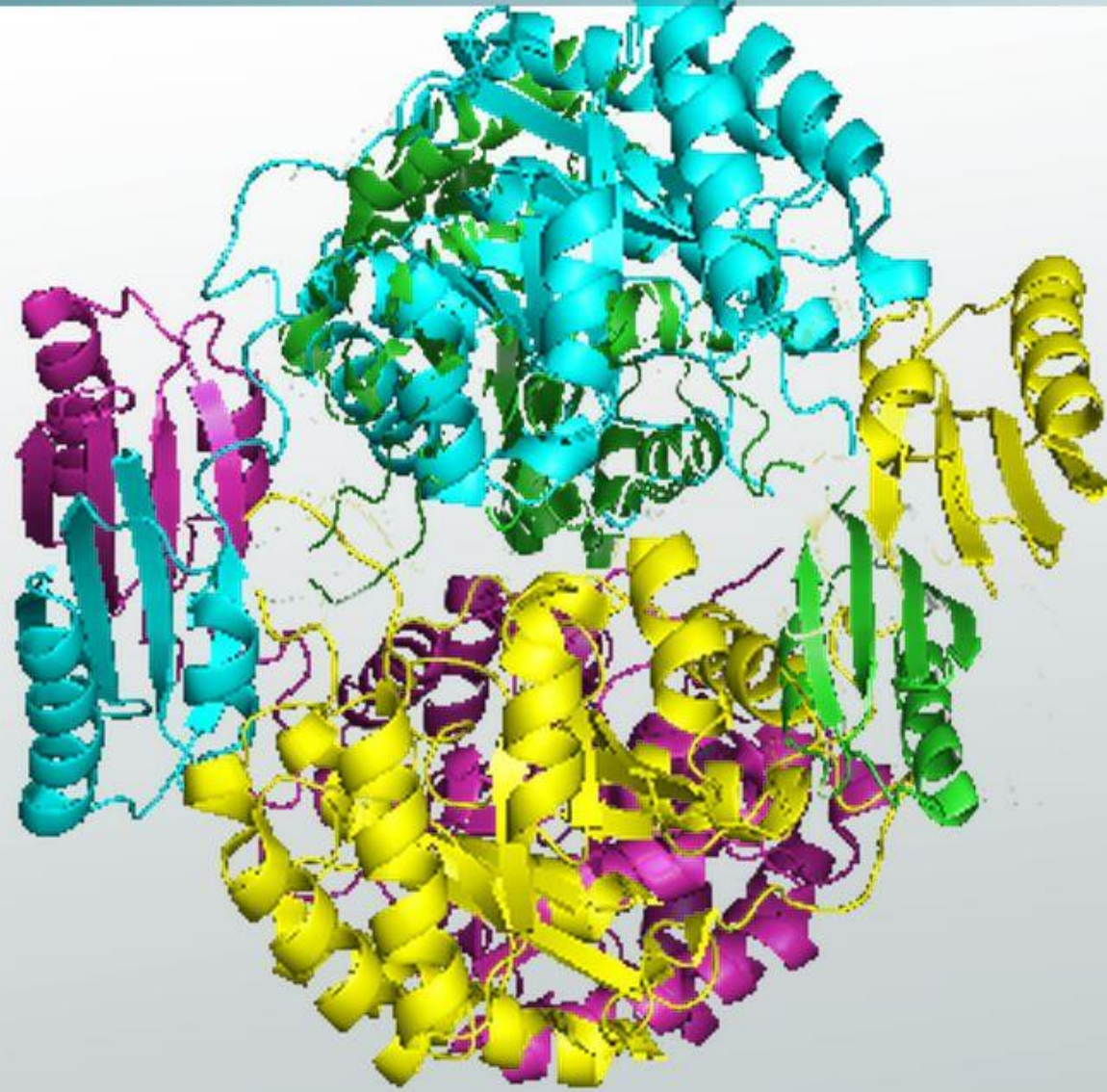
The process by which a polypeptide chain acquires its intricate 3D tertiary structure is called **protein folding**.

- **Spontaneous Folding:** In many cases, the amino acid sequence (primary structure) contains all the information necessary for the protein to spontaneously fold into its correct conformation under physiological conditions.
- **Molecular Chaperones:** For many proteins, especially larger ones or those in crowded cellular environments, specialized proteins called **molecular chaperones** assist in the proper folding process, preventing misfolding or aggregation.
- **Misfolding Diseases:** Incorrect protein folding can lead to protein aggregation and is implicated in many devastating diseases, including Alzheimer's, Parkinson's, and Creutzfeldt-Jakob disease (prion diseases).

Quaternary Structure: Multi-Subunit Assemblies

- Some functional proteins are composed of more than one polypeptide chain. The **quaternary structure** refers to the arrangement of these multiple polypeptide subunits in a multi-subunit protein.
- **Subunits:** Individual polypeptide chains within a multi-subunit protein are called **subunits** or protomers. These subunits can be identical (homooligomeric) or different (heterooligomeric).
- **Stabilization:** The interactions holding subunits together are the **same non-covalent interactions** that stabilize tertiary structure (hydrophobic interactions, hydrogen bonds, ionic bonds) and sometimes **disulfide bonds**.

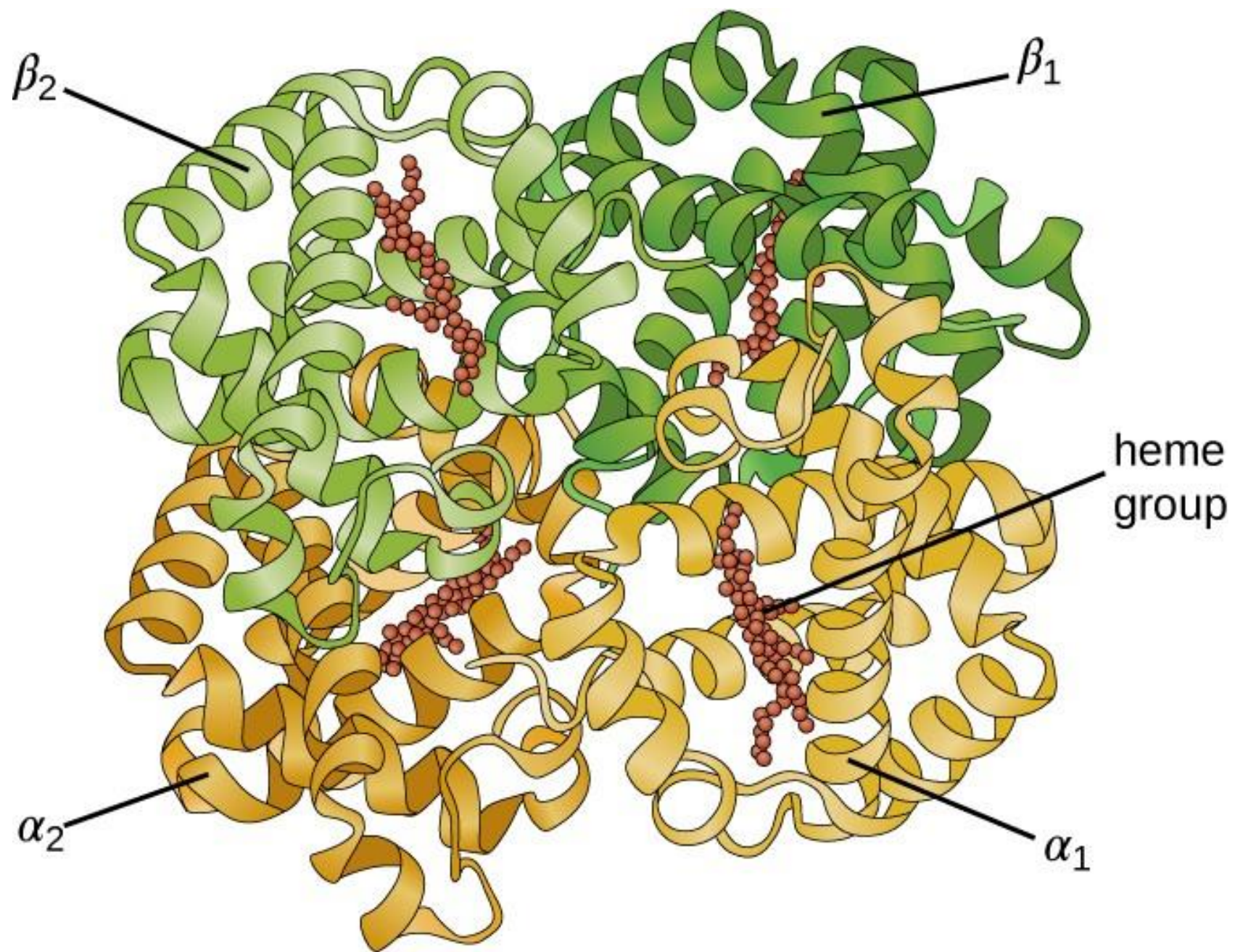
WHAT IS A QUATERNARY PROTEIN STRUCTURE?



Quaternary Structure: Multi-Subunit Assemblies

Examples

- **Hemoglobin:** A classic example. It's a tetramer composed of four polypeptide subunits (two α -chains and two β -chains). Each subunit carries one heme group for oxygen binding. The cooperative binding of oxygen to these subunits is a key feature of hemoglobin function.
- **Insulin Receptor:** A dimer, crucial for cell signaling.
- **RNA Polymerase:** A complex multi-subunit enzyme involved in transcription.



Quaternary Structure: Multi-Subunit Assemblies

Advantages of Quaternary Structure

- **Increased Stability:** Larger complexes can be more stable than individual subunits.
- **Regulatory Mechanisms:** Quaternary structure often provides more sophisticated ways to regulate protein activity (e.g., allosteric regulation, where binding of a molecule to one subunit affects the activity of other subunits).
- **Efficiency:** Multiple active sites can cooperate.
- **Assembly of Complex Functions:** Different subunits can contribute different catalytic or binding activities to a single functional unit.

Diverse Functions of Proteins

Enzymatic Catalysis:

- **Function:** Proteins called **enzymes** are biological catalysts. They dramatically increase the rate of biochemical reactions by lowering the activation energy, without being consumed in the process.
- **Mechanism:** They possess specific **active sites** that bind to particular substrate molecules with high specificity.
- **Examples:** Hexokinase (phosphorylates glucose in glycolysis), DNA polymerase (synthesizes DNA), Amylase (digests starch).

Structural Support:

- **Function:** Provide shape, strength, and elasticity to cells, tissues, and organisms.
- **Examples:**
 - **Collagen:** The most abundant protein in mammals. A fibrous protein that forms strong fibers in connective tissues (skin, bone, tendons, cartilage). Provides tensile strength.
 - **Keratin:** Fibrous protein found in hair, nails, skin, feathers, and horns. Provides protection and structure.
 - **Actin and Tubulin:** Form cytoskeletal filaments in cells, providing structural support, facilitating cell movement, and intracellular transport.

Diverse Functions of Proteins

Transport and Storage:

- **Function:** Bind and carry specific molecules within cells, between cells, or throughout the body.
- **Examples:**
 - **Hemoglobin:** Transports oxygen in red blood cells.
 - **Myoglobin:** Stores oxygen in muscle cells.
 - **Albumin:** Major protein in blood plasma, transports fatty acids, hormones, and drugs.
 - **Membrane Transporters:** Proteins embedded in cell membranes that move ions and molecules across the membrane (e.g., glucose transporters, ion channels).

Immune Defense:

- **Function:** Identify and neutralize foreign invaders (bacteria, viruses) and abnormal cells.
- **Examples:**
 - **Antibodies (Immunoglobulins):** Highly specific proteins produced by B cells that recognize and bind to antigens.
 - **Components of the complement system:** Proteins that enhance the immune response.

Diverse Functions of Proteins

Motion:

- **Function:** Generate movement at the cellular and organismal levels.
- **Examples:**
 - **Actin and Myosin:** Key proteins in muscle contraction.
 - **Dynein and Kinesin:** Motor proteins that move along microtubules, involved in vesicle transport and flagellar/ciliary movement.

Regulation and Signaling:

- **Function:** Mediate communication between cells, tissues, and organs; regulate gene expression and cell growth.
- **Examples:**
 - **Hormones:** Many hormones are peptides or proteins (e.g., Insulin, Glucagon, Growth Hormone).
 - **Receptors:** Proteins on cell surfaces or inside cells that bind to specific signaling molecules (ligands) and initiate a cellular response (e.g., G protein-coupled receptors).
 - **Transcription Factors:** Proteins that bind to DNA and regulate gene expression.

Nutrient Storage:

- **Function:** Store essential nutrients.
- **Examples:**
 - **Ferritin:** Stores iron in cells.
 - **Casein:** Protein in milk, a source of amino acids for offspring.
 - **Ovalbumin:** Protein in egg whites, a nutrient for developing embryos.

Questions

- A research team is investigating the thermal stability of a purified globular enzyme. They measure its activity across a range of temperatures and observe that its activity sharply drops above 55°C, even though the assay conditions (pH, substrate concentration) are kept optimal.

Explain this phenomenon:

- a. What general process is occurring when the enzyme's activity sharply drops at elevated temperatures? Define this process in the context of protein structure.
- b. Which specific non-covalent interactions contributing to the protein's tertiary structure are most likely to be disrupted first by increasing temperature, and why?
- c. If this enzyme were part of a larger, multi-subunit complex, how might the process described in (a) affect the overall complex? Would its subunits necessarily unfold simultaneously and to the same extent? Explain.
- d. Could the enzyme potentially regain its activity if cooled back down from 60°C to 25°C? What factors would influence the likelihood of successful refolding?